JAYRAJ SHARAD PUNEKAR

B.Tech in Mechanical and Automation Engineering (2025 Batch)
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SUMMARY

Aspiring Design Engineer with knowledge of Siemens NX and CATIA V5 for 3D part modeling, surface modeling, and assembly creation. Skilled in close volume creation, B-side feature design, and fundamentals of GD&T and DFM/DFA. Seeking an entry-level opportunity to apply CAD skills and contribute to engineering design projects.

TECHNICAL SKILLS

- **Domain Knowledge:** Automotive Interior Plastic Trim Design, Plastic Injection Molding.
- **Design Knowledge:** Close Volume creation, B-side features, Tooling Direction (MTD), Parting Line & Lifter Basic, Draft Analysis, Power Copy.
- **DFM/DFA:** Basic understanding of Design for Manufacturing and Design for Assembly fundamentals.
- **CAD Tools:**
  CATIA V5: Surface & Solid Modeling, Assembly design, Draft Analysis, and Basic Sheet Metal Design.
  Siemens NX (UG NX): 3D Part Modeling, Assembly Creation, and Drafting.
  AutoCAD: 2D Drafting.
- **GD&T:** Knowledge of geometric dimensioning and tolerancing concepts, symbols, and datum systems.

PROJECT

Motorized Hydraulic Jack

- Modified a 2-ton hydraulic jack by integrating a cam mechanism driven by a 12V DC motor.
- Enabled automatic operation of the jack to lift vehicles with reduced manual effort.

TRAINING

Hands-on Plastic Trim Design – CATIA V5 (4 Months)

- Close Volume creation and Class-B & Class-C surface development
- B-Side Features: Screw bosses, Locators, Doghouses, Snap-fits, Pendulum snaps
- Tooling considerations: Main Tooling Direction (MTD), Parting Line & Lifter basics, Draft Analysis
- Power Copy creation for parametric B-side features and master section interpretation
- Complete workflow: Close Volume + Master Section + B-side feature implementation

INTERNSHIP

Intern: Sheet Metal Department (1Month)
Rahul Electricals, Hotgi Rd, Solapur, Maharashtra

- Assisted in fabrication and mechanical assembly of electrical control panel box.

EDUCATION

B.Tech in Mechanical and Automation Engineering, Walchand Institute of Technology, Solapur (2025)
Grade: 9.43 CGPA

Diploma in Mechanical Engineering, Government Polytechnic, Solapur (2022)
Grade: 79.54% First Class with Distinction

SSC, Siddheshwar High School, Solapur (2019)
Grade: 67.60%

CERTIFICATIONS & ACHIEVEMENT

AutoCAD Certification – CADKNEZ Solapur **(90 Hours)**

- Completed training in 2D drafting and mechanical drawings using AutoCAD.
- Achieved 1st Rank in AutoCAD Government Exam conducted by the CADKNEZ Solapur.

CATIA V5 Certification – CAD Solutions **(120+ Hours)**

- Trained in Part Design, Assembly, Drafting, Sheet Metal, and Surface Design in CATIA V5.

CATIA V5: Interior Surfacing (Basic to Advanced) – Udemy **(30+ Hours)**

- Completed a course on interior surfacing for automotive plastic trims in CATIA V5.